



GOVIND SHAKYA
Bachelor Of Technology
in MATHEMATICS AND COMPUTING
Indian Institute Of Technology, Ropar

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GitHub, LinkedIn
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EDUCATION

Degree	Institute/Board	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	2020-2024
Higher Secondary	State board of Madhya Pradesh	2017-2019

EXPERIENCE

• Specialist Programmer L2

Jul 2024 – Present

Infosys

- Engineering and maintaining full-stack features for the **Roland Garros** website under the Infosys Sports Platform, integrating high-volume data ingestion pipelines using **RabbitMQ** and automated triggers to normalize metadata, ensuring 99% uptime and seamless real-time updates.
- Designed, and deployed secure **RESTful APIs** powering the RG Match Center, enabling interactive match data visualization for 50K+ concurrent users and reducing average response latency by 30%.
- Built a missing-packets analysis platform for AO and RG tournaments, supporting real-time packet loss detection and visualization using **AWS DynamoDB, PostgreSQL, and Amazon S3**, reducing troubleshooting time by 25%.
- Implemented a platform activity tool to migrate production data into lower environments (Dev, Staging, Preprod) via **PostgreSQL bulk COPY** operations, improving synchronization speed by 40% and enhancing environment reliability.
- Optimized Pattern-of-Play analysis queries and AI commentary pipelines (French language), reducing query execution time by 45% and improving multilingual chatbot visualization performance.
- **Technologies:** Java, Spring Boot, AWS (Lambda, S3, DynamoDB), PostgreSQL, Node.js, RabbitMQ, REST APIs.

• Software Development Engineer Intern

May 2023 – Jul 2023

Strand Life Sciences

- Developed a customized API for the **Gene Expression Omnibus (GEO)** Metadata Curation Tool, streamlining large-scale data ingestion and curation workflows, reducing manual processing by 60%.
- Applied advanced **Natural Language Processing (NLP)** techniques using **spaCy** to extract unstructured metadata from NCBI pages, expanding automated retrieval capabilities and improving accuracy by 35%.
- Designed a user-friendly interface with **React** and **Material UI**, enabling researchers to explore, edit, and export curated metadata in **YAML** and **JSON** formats, increasing adoption rate among internal teams.
- **Technologies:** React, Flask, Material UI, NLP (spaCy), MySQL, REST APIs.

PROJECTS

Pharmacy Management System

Sep 2023 – Oct 2023

Personal Project

GitHub

- Devised a full-stack pharmacy management system with **PostgreSQL** backend, covering employee management, drug inventory, sales, customer records, and authentication, supporting 500+ simulated transactions.
- Architected trigger functions for automated employee license validation, drug expiration alerts, and inventory checks, reducing manual errors by 30%.
- Executed role-based access control: administrators perform **CRUD operations** on drugs and employees, while employees manage sales and generate receipts, improving system security and usability.
- **Technologies:** React.js, Node.js, PostgreSQL, Chakra UI.

ML Research Intern (Gaussian Process Regression for Option Pricing)

Jan 2024 – Apr 2024

Under Dr. Puneet Pasricha (IIT Ropar)

- Researched **Gaussian Process Regression (GPR)** to accelerate option pricing tasks and Greek estimation for advanced financial models, reducing runtime by up to 60%.
- Demonstrated significant speed-ups for fitting Greek profiles and summarizing implied volatility surfaces, with minor accuracy trade-offs, enabling faster quantitative analysis.
- Applied scikit-learn pipelines to price options under complex models and approximate volatility surfaces for downstream use in quantitative finance.
- **Technologies:** Python, scikit-learn, Gaussian Process Regression, Quantitative Finance.

Real Time Chat App

May 2022 – Jul 2022

Personal Project

GitHub

- Delivered a scalable real-time chat platform using **React, Node.js, Express, Socket.io** with MongoDB for persistence and Redux for state management, supporting 1K+ concurrent users in testing.
- Crafted features such as one-to-one and group messaging, push notifications, group admin controls, and role-based group permissions, improving collaboration efficiency.
- Handled edge cases including offline message queuing, read receipts, and notification throttling, improving reliability and user experience by 40%.
- **Technologies:** React.js, Node.js, Express.js, Socket.io, MongoDB, Chakra UI.

TECHNICAL / PROGRAMMING SKILLS

- **Programming Languages:** C++, Java, JavaScript, Python.
- **DevOps Platform:** AWS Bedrock, AWS Lambda, AWS CloudFront.
- **WebDev Frameworks/Libraries:** SpringBoot, Angular, React, NodeJs, Flask.
- **Version Control and Collaboration:** Git, GitHub.